SPRING 2026

Nicole Marcial

ITAI 2376 Deep Learning Portfolio

Portfolio focused on applying deep learning concepts through hands-on labs and a working cookie recipe adaptation agent.

GitHub: https://github.com/2002nicole

Portfolio: https://github.com/2002nicole/Nicole-Marcial-DeepLearning-ITAI2376

Final Project: https://github.com/2002nicole/NicoleMarcial_Solo_ITAI2376

Learning Journey : From Neural Networks to AI Agents

Modules 02–03 Neural Network Foundations

MLPs, CNNs, image classification, transfer learning s

Modules 04–05 Sequence + Language Models

RNNs, NLP, transformers, attentio

Modules 06–08 Generative AI

VAEs, GANs, U-Net, diffusion models, CLIP evaluation

Modules 09–12 AI Agents + Reasoning

RL, RLHF, RAG, LangChain, tools, single-agent design

Final Project: Cookie Recipe Adaptation Agent

LangChain | RAG | Custom Tools | Gradio

What I practiced

- Training and evaluation
- Reading loss/accuracy curves
- Comparing model behavior
- Debugging Colab notebooks

Frameworks used

- PyTorch
- TensorFlow/Keras
- LangChain
- Chroma + embeddings
- Gradio

Featured Project: Cookie Recipe Adaptation Agent

A single AI agent that helps beginner bakers adapt cookie recipes through substitutions, batch scaling, dietary changes, and texture guidance.

User request → **LLM Agent Brain** → **Custom Tools** → **Retrieved Baking Knowledge** → **Adapted Recipe Output**

Custom Tools

- Recipe Retrieval Tool
- Ingredient Scaling Tool
- Substitution Guidance Tool
- Recipe Adaptation Tool

Project repo:
https://github.com/2002nicole/NicoleMarcial_Solo_ITAI2376

Skills & Competencies

Deep Learning Skills

- Neural network training and evaluation
- CNN image classification
- Transfer learning
- Diffusion model concepts
- Reinforcement learning foundations
- Transformer-based language models

Professional Skills

- Project scoping
- Debugging and testing
- Explaining technical results clearly
- Writing reproducible documentation
- Connecting AI tools to real user needs

Reflection & Next Steps

Reflection

This course helped me move from running guided notebooks to building a focused AI agent with retrieval, custom tools, and a demo interface.

Next steps

Improve the cookie agent with stronger memory, better unit conversion, a larger baking knowledge base, and more personalized user preferences.

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